

67. Add Binary

Easy

Topics

Companies

Given two binary strings `a` and `b`, return *their sum as a binary string*.

Example 1:

Input: `a = "11", b = "1"`
Output: `"100"`

Example 2:

Input: `a = "1010", b = "1011"`
Output: `"10101"`

Constraints:

- `1 <= a.length, b.length <= 104`
- `a` and `b` consist only of `'0'` or `'1'` characters.
- Each string does not contain leading zeros except for the zero itself.

Python:

class Solution:

```
def addBinary(self, a: str, b: str) -> str:
    return bin(int(a, 2) + int(b, 2))[2:]
    # result = []
    # carry = 0

    # i, j = len(a) - 1, len(b) - 1

    # while i >= 0 or j >= 0 or carry:
```

```

#   total = carry
#   if i >= 0:
#       total += int(a[i])
#       i -= 1
#   if j >= 0:
#       total += int(b[j])
#       j -= 1

#   result.append(str(total % 2)) # store the current bit
#   carry = total // 2           # update carry

# return ".join(reversed(result))

```

JavaScript:

```

/**
 * @param {string} a
 * @param {string} b
 * @return {string}
 */
var addBinary = function(a, b) {
    let i = a.length - 1;
    let j = b.length - 1;
    let carry = 0;
    let result = [];

    while (i >= 0 || j >= 0 || carry > 0) {
        let sum = carry;

        if (i >= 0) sum += Number(a[i--]); // add digit from a
        if (j >= 0) sum += Number(b[j--]); // add digit from b

        result.push(sum % 2); // current bit
        carry = Math.floor(sum / 2); // update carry
    }

    return result.reverse().join("");
};

```

Java:

```

class Solution {
    public String addBinary(String a, String b) {
        StringBuilder result = new StringBuilder();

        int i = a.length() - 1;
        int j = b.length() - 1;
    }
}

```

```
int carry = 0;

// Traverse both strings from right to left
while (i >= 0 || j >= 0 || carry > 0) {
    int sum = carry;

    if (i >= 0) {
        sum += a.charAt(i) - '0'; // convert char to int
        i--;
    }
    if (j >= 0) {
        sum += b.charAt(j) - '0';
        j--;
    }

    // Append current bit
    result.append(sum % 2);

    // Update carry
    carry = sum / 2;
}

// Reverse result because we added from right to left
return result.reverse().toString();
}
```